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System for supplying a filling to foodstuff products

Abstract

A system for supplying a filling to foodstuff products, including a conveyor for feeding along a path of advance a succession of rows of foodstuff products oriented in a direction transverse to the path of advance, a dispensing head for dispensing filling onto or into the products of each row of the succession of foodstuff products rows, and a vibrating station for subjecting each conveyor-fed row of products to an oscillatory motion to spread out the filling dispensed. The at least one conveyor receives the products in a free state, where they rest on a conveying surface without constraint in any specific position. The system includes a containment unit above the conveyor configured for containing the rows of products while being subjected to oscillatory motion by the vibrating station.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims the benefit of Italian Patent Application No. 102024000004990, filed on Mar. 6, 2024, the disclosures of which are herein incorporated by reference in its entirety.

[0002] The present invention regards a system for supplying a filling to foodstuff products, the system comprising: [0003] at least one conveyor for feeding along a path of advance a succession of rows of foodstuff products oriented in a direction transverse to the path of advance; [0004] a dispensing head for dispensing the filling onto or into the products of each row of the succession of rows of foodstuff products; and [0005] a vibrating station for subjecting each row of products fed by said at least one conveyor to an oscillatory motion to spread out the filling dispensed.

[0006] A system of the type described above may for example be used for dispensing a filling on oven-baked products, such as biscuits. Via the above system, the filling is dispensed onto the biscuit, for example within a cavity made therein, and then the filling is spread out so that it comes to cover the entire cavity as a result of the oscillatory motion caused by the vibrating station.

[0007] In solutions according to the prior art, the conveyor configured for feeding the products through the vibrating station comprises a plurality of supports provided with housings for receiving the rows of products and keeping them in the respective orderly arrangement during the action exerted thereon by the vibrating station.

[0008] The above solutions according to the prior art present, however, the drawback of requiring replacement of the aforesaid supports of the conveyor with new supports of appropriate shape, whenever there is envisaged start of operation of the system on a new type of foodstuff products.

[0009] In this context, the object of the present invention is to provide a system of the type referred to at the start that will be improved over known solutions.

[0010] In particular, an object of the present invention is to provide a more versatile system.

[0011] A further object of the present invention is to provide a system capable of achieving larger production volumes.

[0012] One or more of the aforesaid objects are achieved via a system according to claim **1**.

[0013] The present invention moreover regards a method according to claim **11**.

[0014] The claims form an integral part of the teaching provided herein.

Description

[0015] Further characteristics and advantages of the present invention will emerge clearly from the ensuing description with reference to the annexed drawings, which are provided purely by way of non-limiting example and in which:

[0016] FIG. **1** is a schematic illustration of an example of the system described herein; and

[0017] FIG. **2** illustrates, according to a top plan view, a part of the system of FIG. **1**.

[0018] In the ensuing description, various specific details are illustrated aimed at enabling an in-depth understanding of the embodiments. The embodiments may be obtained without one or more of the specific details, or with other methods, components, materials, etc. In other cases, known structures, materials, or operations are not illustrated or described in detail so that various aspects of the embodiment will not be obscured.

[0019] The references used herein are provided merely for convenience and hence do not define the sphere of protection or the scope of the embodiments.

[0020] As anticipated above, the present invention regards a system for supplying a filling to foodstuff products.

[0021] It should be noted that this solution has been devised specifically for applications on oven-baked products, in particular biscuits, but can in any case be advantageously used also for other types of foodstuff products, for example confectionery products, such as pralines or filled wafers, ice-creams, semifreddo, products of the pasta industry, products of the dairy sector, etc.

[0022] With reference to FIGS. 1 and 2, the system described herein—designated as a whole by the reference number **10**—comprises a conveyor **20** for feeding along a path of advance K a succession of rows **100** of foodstuff products **101** oriented in a direction transverse to the path of advance K.

[0023] According to an important characteristic of the system described herein, the conveyor **20** is configured for receiving the products **101** in a free state where these rest simply on a conveying surface, without being constrained in any way in a specific position.

[0024] In various preferred embodiments like the one illustrated, the conveyor **20** comprises, in succession, a first conveyor belt **21A**, a second conveyor belt **21B**, and a third conveyor belt **21C**, which have respective conveying surfaces **22A**, **22B**, **22C** that extend along the path of advance K.

[0025] The conveyor belt **21A** is configured for receiving on its conveying surface **22A** the rows of products **100** one after another from a station located upstream.

[0026] The system **10** further comprises a dispensing head **40** for dispensing a filling onto or into the products **101** of each row **100** that is positioned on top of the conveying surface **22A** of the conveyor belt **21A**.

[0027] In the example illustrated, the products **101** have a cavity **101A** that is to receive the filling dispensed by the dispensing head **40**. It should be noted that not necessarily do the foodstuff products on which the system described herein operates have to have a cavity for receiving the filling; they may, in fact, alternatively have just one surface that is to be coated with the filling.

[0028] The dispensing head **40** may be of any known type suited for the purposes referred to.

[0029] Preferably, the filling used is a viscous fluid material, for example characterized by a creamy consistency.

[0030] Downstream of the dispensing head **40** and in the region of the second conveyor belt **21B**, the system **10** comprises a vibrating station **60** configured for subjecting each row of products **100** arranged on the conveying surface **22B** of the conveyor belt **21B** to an oscillatory motion so as to spread out the filling previously dispensed on the products **101**.

[0031] In this connection, with reference to FIG. 2, it may be noted how the filling of the products **101** of the row **100(I)**, which has just passed underneath the dispensing head **40**, is basically concentrated in a central region of the cavity **101A** of the products **101**, and how, instead, the filling of the products **101** of the row **100(II)**, which has passed beyond the vibrating station **60**, is spread out and substantially coats the entire cavities **101A**.

[0032] In general, the action exerted by the vibrating station **60** has the function of spreading out the filling on the products so that it comes to coat a surface of these wider than the surface coated just after it has been dispensed by the dispensing head **40**.

[0033] Preferably, the vibrating station **60** is operatively connected to at least a stretch of the conveying surface **22B** of the conveyor belt **21B** so as to subject the stretch itself to an oscillatory motion that will then involve the rows of products **100** that are passing through the stretch in question.

[0034] The person skilled in the sector will understand that the vibrating station **60** may extend even over the entire conveying surface **22B** of the second conveyor belt **21B**, in which case the vibrations generated by the station **60** propagate as far as the two opposite ends of the surface.

[0035] The vibrating station **60** may itself also be of a known type suited for the purposes referred to.

[0036] The third conveyor belt **21C** receives the rows of products **100** from the second conveyor belt **21B** to feed them to the subsequent stations of the production line; the respective products **101**

present the filling spread out as required.

[0037] According to a further important characteristic of the system described herein, this comprises a containment unit **80** configured for containing the rows of products **100** while these traverse the vibrating station **60** and are subjected to the oscillatory motion referred to above.

[0038] In one or more preferred embodiments like the one illustrated, the containment unit **80** is positioned above the conveyor **20** and in the region of the vibrating station **60**.

[0039] Preferably, the containment unit **80** extends from a position **P1** upstream of the vibrating station **60**, with reference to the path of advance K, and as far as a position **P2** downstream of the vibrating station **60**.

[0040] Even more preferably, the position **P1** is in the region of the first conveyor belt **21A**, whereas the position **P2** is in the region of the third conveyor belt **21C**.

[0041] In this way, as will be seen more clearly in what follows, the rows of products **100** are engaged and released by the containment unit **80** in positions in which they are certainly not subjected to the vibrations generated by the vibrating station **60**. In alternative embodiments in which the vibrating station **60** is limited to a stretch of the conveyor belt to which it is associated, for example through the use of damping or isolating means capable of limiting propagation of the vibrations, the two positions **P1** and **P2**, or at least one of these, may be located in the region of the conveyor belt in question. The person skilled in the sector will hence understand that the use of three conveyor belts as in the example illustrated does not constitute an essential characteristic.

[0042] In one or more preferred embodiments like the one illustrated, the containment unit **80** comprises a guide **82** that extends along a closed-loop path and has a preferably rectilinear bottom stretch **82A**, a first, preferably curvilinear, stretch of reversal of motion **82B**, a preferably rectilinear top stretch **82C**, and a second, preferably curvilinear, stretch of reversal of motion **82D**.

[0043] In one or more preferred embodiments like the one illustrated, the bottom stretch **82A** extends along the path of supply K starting from the position **P1** and as far as the position **P2**.

[0044] In one or more preferred embodiments like the one illustrated, the containment unit **80** further comprises a plurality of units **84** mobile along the guide **82**, which each carry a respective containment member **90**.

[0045] The containment members **90** have the function of providing circumscribed spaces S individually housed within which are the products **101** of the rows of products **100** that advance through the vibrating station **60** along the path of advance K.

[0046] In one or more preferred embodiments like the one illustrated, each containment member **90** comprises a beam **92** set in a direction parallel to the conveyor belt **21** and oriented in a direction transverse to the path of advance K. Moreover, the containment member **90** comprises a series of tabs **94**, all projecting from the beam **92** on one and the same side, which are set facing the conveying surface **22B** of the conveyor belt **21B** when the containment member **90** and the respective mobile unit **84** are located along the bottom stretch **82A** of the guide **82**.

[0047] The tabs **94** define containment surfaces **94A** configured for delimiting the circumscribed spaces S.

[0048] In one or more preferred embodiments like the one illustrated, the tabs **94** of one and the same containment member **90** define with their own containment surfaces **94A** a series of V-shaped profiles oriented in the direction of the path of advance K or in the opposite direction. The usefulness of the V-shaped profiles will become evident in what follows.

[0049] The series of containment members **90** provides for a continuous alternation of one containment member **90** with containment surfaces **94** that define V-shaped profiles oriented in the direction of the path of advance K and one containment member **90** with containment surfaces **94** defining V-shaped profiles oriented in the opposite direction.

[0050] During operation of the system, the mobile units **84** with the respective containment members **90** move along the guide **82** so that between the position **P1** and the position **P2** along the path of advance K each row of products **100** is accompanied by a containment member **90(I)**,

which sets itself downstream of the row of products **100**, along the path of advance K, and by a further containment member **90(II)**, which sets itself upstream of the same row, along the path of advance K.

[0051] The pair of the two containment members **90(I)**, **90(II)** delimits the aforesaid circumscribed spaces S.

[0052] The stretch **82A** enables the mobile units **84** to accompany with the respective containment members **90** the rows of products along the path of advance K. The stretches **82B**, **82C**, **82D** of the guide **82** enable the mobile units **84** to bring back their respective containment members **90** to the position P1 to operate on new rows of products coming from the dispensing head **40**.

[0053] In one or more preferred embodiments like the one illustrated, the system **10** comprises a linear electric motor **86** that develops along the guide **82** and is configured for movement of the mobile units **84** along the guide itself (in FIG. **1** the motor **86** is illustrated only partially for simplicity of representation).

[0054] In particular, in one or more preferred embodiments like the one illustrated, the electric motor **86** comprises a series of electrical windings **86A** arranged along the guide **82** and a plurality of magnets **86B** fixed to the mobile units **84** and configured for interacting electromagnetically with the windings **86A** to bring about a movement of the mobile units **84** along the guide **82**.

[0055] Electric motors of this type are in themselves already known in the art, for example from the document EPO 939 482B1, and afford the advantage of providing a highly versatile movement system in which each of the mobile units can be moved in a way independent of the others; in particular, the movement parameters of each mobile unit, such as the position, speed, and acceleration, can be set in a way independent of the other mobile units.

[0056] According to an alternative embodiment, instead of the electric motor **86** described above, the system **10** may comprise a plurality of electric motors, each set on a respective mobile unit **84** for driving movement thereof.

[0057] In one or more preferred embodiments like the one illustrated, the system comprises a control unit **70** configured for driving the electric motor **86**.

[0058] In one or more preferred embodiments, the control unit **70** is configured for driving the electric motor **86** as a function of a signal indicating the position of the rows of products **100** along the path of advance K, in a stretch upstream of the containment unit **80**.

[0059] For this purpose, the system may for example comprise a camera **72** set above the first conveyor belt **21A** and configured for detecting the rows of products **100** that advance along the path of advance K.

[0060] The control unit **70** or some other control unit of the system can use the data received from the camera **72** to generate data indicating the position of the rows of products **100** along the path of advance K.

[0061] The system may in any case use other types of devices for detecting the position of the rows of products **100**.

[0062] With reference now to operation of the system **10**, this envisages that a plurality of rows of products **100** will be supplied one after another to the conveyor belt **21A**.

[0063] The dispensing head **40** operates for dispensing the filling within the cavity **101A** of the products **101**, as the rows of products **100** move underneath the head itself.

[0064] Incidentally, it should be noted that the dispensing head **40** may be configured to operate at the same time on just one row of products or else on a plurality of rows of products.

[0065] Simultaneously, the conveyor belt **21A** can be operated to bring about a continuous movement of the rows of products **100** or else a step-like movement thereof. In the latter case, the dispensing head **40** can be moved according to a movement of tracking of the rows of products **100** along the path of advance K.

[0066] The rows of products **100** on which the filling has been dispensed then move, as a result of their advance, onto the conveying surface **22B** of the second conveyor belt **21B**.

[0067] With reference to FIGS. 1 and 2, before a single row of products **100** is transferred from the first conveyor belt **21A** to the second conveyor belt **21B**, its products **101** are received within the circumscribed spaces S defined by a pair of containment members **90** of the containment unit **80**. [0068] In particular, immediately before the row of products **100** reaches the position **P1**, a containment member **90(I)** is arranged by the respective mobile unit **84(I)** on the path of advance K, downstream of the row of products **100**, and immediately after the same row **100** has passed beyond the position **P1**, a further containment member **90(II)** is arranged by the respective mobile unit **84(II)** on the path of advance K, upstream of the row **100**.

[0069] Incidentally, it is pointed out that the first stretch of reversal of motion **82D** enables approach of the two containment members **90(I)**, **90(II)** to the conveying surface **22B** and to the row of products **100**, according to a movement of rotation from above downwards that facilitates positioning of the two containment members **90** up against the row of products **100** without interfering with it.

[0070] After the row of products **100** has been enclosed in the circumscribed spaces S defined by the two containment members **90(I)**, **90(II)**, these, carried by the two mobile units **84(I)**, **84(II)**, move along the path of advance K substantially at the same speed of advance as that of the row of products **100** so as to accompany it during its advance and keep its products within the circumscribed spaces S.

[0071] At this point, the row of products **100** is transferred onto the second conveyor belt **21B** and subsequently passes through the vibrating station **60**.

[0072] When the row of products **100** is subjected to the oscillatory motion discussed above in the vibrating station **60**, the containment members **90** perform the function of keeping the products **101** within the circumscribed spaces S, preventing them from shifting in an uncontrolled manner from their predefined positions.

[0073] In this connection, with reference to FIG. 2, it may be noted that between the containment surfaces **94A** of the containment members **90** and the products **101** there may be a certain gap on account of which the aforesaid oscillatory motion may slightly displace the products **101** from their predefined positions, but in any case in a controlled manner, once again within the limits of the circumscribed spaces S.

[0074] Preferably, starting from a position **P3** downstream of the vibrating station **60** and upstream of the position **P2**, the containment member **90(I)** is moved by the respective mobile unit **84(I)** at a speed lower than the speed of advance of the row of products **100** so that a relative displacement is obtained between the containment member **90(I)** and the products **101** such as to bring the latter to bear upon the containment member itself and bring about an alignment thereof in one and the same direction transverse to the path of advance K.

[0075] It may now be noted that the V-shaped profiles defined by the containment surfaces **94A** are able to bring about, as a result of the relative displacement between the containment member **90(I)** and the products **101**, centering of the products on a plurality of reference axes K_i parallel to the path of advance K, in addition to the aforesaid alignment of the products in a direction transverse to the path of advance.

[0076] Preferably, the position **P3** is already located in a region of the third conveyor belt **21C**, downstream of the conveyor belt **21B**. The position **P3** may in any case also be located in a region of the second conveyor belt **21B**, especially in the cases where the propagation of the vibrations along the conveying surface **22B** thereof is limited to a circumscribed stretch. In an alternative embodiment, the alignment and centering of the products **101** described above may be obtained in an altogether similar way through a movement of the mobile unit **84(II)** and of the respective containment member **90(II)** at a speed higher than the speed of advance of the products **101**, equally bringing about a relative displacement between the containment member and the products.

[0077] Of course, the operations referred to above for alignment and centering of the products **101** are not to be deemed essential, above all in the cases where the containment members **90** set

themselves without any gap with respect to the products **101** or else with a negligible gap so that the products **101** are in effect fixed in position on the conveying surface **22B**.

[0078] In any case, it is also possible to envisage an alignment system of a known type configured to operate in a region of the conveyor **20** downstream of the containment unit **80**.

[0079] When each of the two containment members **90(I)**, **90(II)** reaches the position **P2**, it frees the row of products **100** through a movement of rotation from below upwards, caused by the advance of the respective mobile unit **84(I)**, **84(II)** along the second stretch of reversal of motion **82B** of the guide **82**. Also in this case, it may be noted that the movement from below upwards induced by the stretch of reversal of motion **82B** facilitates separation of the containment members **90** from the respective row of products **100**, without any risk of interference with the latter.

[0080] The row of products **100** freed can then be supplied to subsequent processing stations, arranged downstream of the system **10** described herein.

[0081] On account of the foregoing, the row of products **100** will be characterized by a perfectly orderly arrangement of the products.

[0082] The person skilled in the sector will note that it is sufficient to configure the control unit **70** for driving the electric motor **86** in an appropriate way in order to reproduce the mode of operation described above of the containment unit **80**.

[0083] Furthermore, in the case where there is the need to start operation of the system on different foodstuff products, it is sufficient to set the control unit **70** in such a way that it brings about a movement of the mobile units **84** useful for obtaining, through the containment members **90**, circumscribed spaces **S** of dimensions suitable for the new products.

[0084] In this connection, it is again pointed out that the V-shaped profiles defined by the containment surfaces **94A** of the containment members **90** enable creation of circumscribed spaces **S** that can be readily adapted to different product shapes, it being always possible to guarantee in any case an effective action of containment by the surfaces themselves.

[0085] In view of the foregoing, it hence appears clearly that the system described herein is able to operate on foodstuff products of different sizes and/or types without any need to modify or replace components of the system, unlike what occurs with the solutions according to the prior art described at the start of the present treatment.

[0086] Of course, without prejudice to the principle of the invention, the details of construction and the embodiments may vary, even significantly, with respect to what has been illustrated herein purely by way of non-limiting example, without thereby departing from the scope of the invention, as defined by the annexed claims.

Claims

1. A system for supplying a filling to foodstuff products, the system comprising: at least one conveyor (**20**) for feeding along a path of advance (**K**) a succession of rows (**100**) of foodstuff products (**101**) oriented in a direction transverse to the path of advance (**K**); a dispensing head (**40**) for dispensing the filling onto or into the products of each row (**100**) of the succession of rows of foodstuff products; and a vibrating station (**60**) for subjecting each row of products fed by said at least one conveyor (**20**) to an oscillatory motion to spread out the filling dispensed, said system being characterized in that: said at least one conveyor (**20**) is configured, at least in part, for receiving the products in a free state, where they simply rest on a conveying surface and are not constrained in any way in a specific position; and said system comprises a containment unit (**80**), positioned above said at least one conveyor (**20**) and configured for containing the rows of products (**100**) while these are subjected to the oscillatory motion by said vibrating station (**60**).

2. The system according to claim 1, wherein said containment unit (**80**) comprises a series of containment members (**90**) mobile along the path of advance (**K**) and configured for defining a plurality of circumscribed spaces (**S**) individually housed within which are the products (**101**) of

said rows of products (100) while these are subjected to said oscillatory motion by said vibrating station (60).

3. The system according to claim 1, wherein said containment unit (80) comprises at least one guide (82), which follows at least in part the path of advance (K) of said at least one conveyor (20), and a plurality of units (84) mobile along said guide (82), which each carry a respective containment member (90) of said series of containment members (90).
4. The system according to claim 3, comprising a linear electric motor (86) that develops along said guide (82) and is configured for moving said mobile units (84) along said guide (82).
5. The system according to claim 4, wherein said linear electric motor (86) comprises a series of electrical windings (86A) arranged in succession along said guide (82) and a plurality of magnets (86B) fixed to said plurality of mobile units (84) and configured to interact electromagnetically with said electrical windings (86A) to bring about a movement of said mobile unit (84) along said guide (82).
6. The system according to claim 1, comprising a control unit (70) configured to control the movement of said series of containment members (90) so that each row of products that traverses said vibrating station (60) is accompanied by a containment member (90(I)) of said series of containment members (90), which is set downstream of said row of products along said path of advance (K), and by a further containment member (90(II)) of said series of containment members (90), which is set upstream of said row of products (100) along said path of advance (K), said containment member and said further containment member (90(I), 90(II)) defining circumscribed spaces (S) individually housed within which are the products of said row (100).
7. The system according to claim 6, wherein said control unit (70) is configured to move said containment member (90(I)) and said further containment member (90(II)) substantially at the same speed as that of said row of products (100) fed by said at least one conveyor (20).
8. The system according to claim 7, wherein said control unit (70) is configured to move said containment member (90(I)) or said further containment member (90(II)) at a speed lower or higher than the speed of said row of products fed by said at least one conveyor (20), after said row has been subjected to said oscillatory motion in said vibrating station (60), so as to obtain a relative displacement between said containment member (90(I)) or said further containment member (90(II)) and the products (101) of said row (100) such as to bring the latter to bear upon said containment member (90(I)) or said further containment member (90(II)) and bring about an alignment thereof in one and the same direction transverse to the path of advance (K).
9. The system according to claim 1, wherein each containment member (90) of said series of containment members (90) has containment surfaces (94A) configured for delimiting said circumscribed spaces (S) that define V-shaped profiles oriented in the direction of the path of advance or in the opposite direction.
10. The system according to claim 1, wherein each containment member (90) of said series of containment members (90) comprises a beam (92) set in a direction parallel to said at least one conveyor and oriented in a direction transverse to the path of advance (K), and a series of tabs (94) all projecting on one and the same side of said beam (92) and defining containment surfaces (94A) configured to delimit said circumscribed spaces (S).
11. A method for supplying a filling to foodstuff products (101), the method comprising: via at least one conveyor (20), feeding along a path of advance (K) a succession of rows of foodstuff products (101) oriented in a direction transverse to the path of advance (K); via a dispensing head (40), dispensing the filling onto or into the products of each row of the succession of rows of foodstuff products; and via a vibrating station (60), subjecting each row of products (100) fed by said at least one conveyor (20) to an oscillatory motion to spread out the filling dispensed; wherein feeding the succession of rows of foodstuff products (100) via said at least one conveyor (20) includes receiving on said conveyor the products in a free state, where they simply rest on a conveying surface and are not constrained in any way in a specific position; said method including, via a

containment unit (80) positioned above said at least one conveyor (20), containing the rows of products (100) while these are subjected to the oscillatory motion by said vibrating station (60).

12. Method according to claim 11, wherein containing the rows of products (100) includes individually housing the products (101) of the rows of products (100) in a plurality of circumscribed spaces (S), delimited by a series of containment members mobile along the path of advance (K), of said containment unit (80).

13. The method according to claim 12, wherein individually housing the products (101) of the rows of products (100) in a plurality of circumscribed spaces (S) includes accompanying each row of products (100) that advances through said vibrating station (60) via a containment member (90(I)) that sets itself downstream of said row of products (100) along said path of advance (K), and via a further containment member (90(II)) that sets itself upstream of said row of products (100) along said path of advance (K), said containment member and said further containment member (90(I), 90(II)) defining circumscribed spaces (S) individually housed within which are the products of said row (100).

14. The method according to claim 13, wherein accompanying each row of products (100) that advances through said vibrating station (60) includes moving said containment member (90(I)) and said further containment member (90(II)) along said path of advance (K) substantially at the same speed as that of said row of products (100) that advances along said path of advance (K).

15. The method according to claim 13, comprising, after said row has been subjected to said oscillatory motion by said vibrating station (60), moving said containment member (90(I)) or said further containment member (90(II)) at a speed lower or higher than the speed of said row of products that advances along said path of advance (K) so as to obtain a relative displacement between said containment member or said further containment member (90(I), 90(II)) and the products of said row such as to bring the latter to bear upon said containment member or said further containment member (90(I), 90(II)) and bring about an alignment thereof in one and the same direction transverse to the path of advance (K).

16. The method according to claim 15, comprising, during said relative displacement between said containment member (90(I)) or said further containment member (90(II)) and the products of said row, engaging said products of said row via containment surfaces (94A) of said containment member or of said further containment member (90(I), 90(II)), which define V-shaped profiles oriented in the direction of the path of advance (K) or in the opposite direction, to bring about centering of said products of said row on a plurality of reference axes parallel to said path of advance (K).
